

Varactor Diode

In this article, we will be discussing about varactor diode. We will also discuss its symbol, construction, its working, and its application in various tuning circuits. In addition to this, we will discuss the characteristic curve of this diode. Later in the article, we will discuss the advantages and disadvantages and Applications of using a varactor diode.

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What is a Varactor Diode?

A varactor diode, also known as a Varicap or volt-cap, is a type of [PN junction diode](#) primarily utilized in the reverse-biased mode. It is a device whose capacitance varies with the variation in the applied reverse bias potential. The term “Varicap” originates from the fusion of the words “variable” and “capacitor.”

While standard diodes are used for current conduction, varactor diodes are specialized for their capacitance characteristics and are primarily operated in reverse bias to exploit their capacitance properties effectively.

Its ability to change its capacitance with applied voltage makes it valuable in oscillator circuits and various tuning applications. Varactor diodes share a similar basic structure with p-n diodes, but their key feature lies in their nonlinear reactance characteristics. This makes them useful for applications where precise tuning or voltage-controlled capacitance is required.

The Vari-cap was first developed by Pacific Semiconductor, a subsidiary of the Ramo Wooldridge Corporation, which obtained . Subsequently, the device was trademarked as the ” Vari-cap” by TRW Semiconductors, the successor to Pacific Semiconductors, in October 1967. These historical facts shed light on the various names associated with the device as it became more widely used.

Symbol of Varactor Diode



symbol of varactor diode

The symbol of a varactor diode bears a resemblance to that of a standard diode. However, it distinguishes itself by representing the cathode terminal using two plates, akin to those found in a capacitor.

Formula of Varactor Diode

$$C_T = \epsilon A / W$$

C_T :- transition capacitance ϵ :-dielectric constant

A :-capacitor's plate area W :-depletion layer's width.

The above relationship shows that transition capacitance is inversely related to depletion layer width. As a result, if we desire a large capacitance magnitude, the width should be minimal. And if we use a low reverse voltage, the width will be minimal. We can simplify it to

$$C_T = C_K / (V_b - V)^m$$

C_T :-transition capacitance V :-applied voltage

C :-diode capacitance when the device is unbiased V_b :-barrier voltage at the junction

m :-constant depending upon the material

$$Q = F / f$$

Q :-Quality factor of the Varactor Diode F :-maximum operating frequency

f :-the operating frequency

Construction of Varactor Diode

The construction of a varactor diode is similar to a regular p-n junction diode, with some additional characteristics. These are added to ensure it shows characteristics of variable capacitance.

The integral part of Varactor diodes is typically made from [semiconductor](#) materials like silicon (Si) or gallium arsenide (GaAs). Silicon varactors are commonly used because they have a lower capacitance range.

A varactor diode typically consists of a p-n junction. P-N junctions are created by adding impurities to semiconductor wafer (doping). The p-type region contains holes (i.e positively charged carriers) and the n-type region contains electrons(negatively charged carriers). The p-n junction creates a depletion region at the boundary between the p-type and n-type regions. The width of this region depends on the kind of biasing technique used in the diode. When forward bias is applied this depletion region will be narrow and when reverse bias is applied the depletion region will be wide blocking most current flow. We also have metal contacts added to the p-type and n-type regions to provide electrical connections. These contacts allow you to apply an external voltage across the diode.

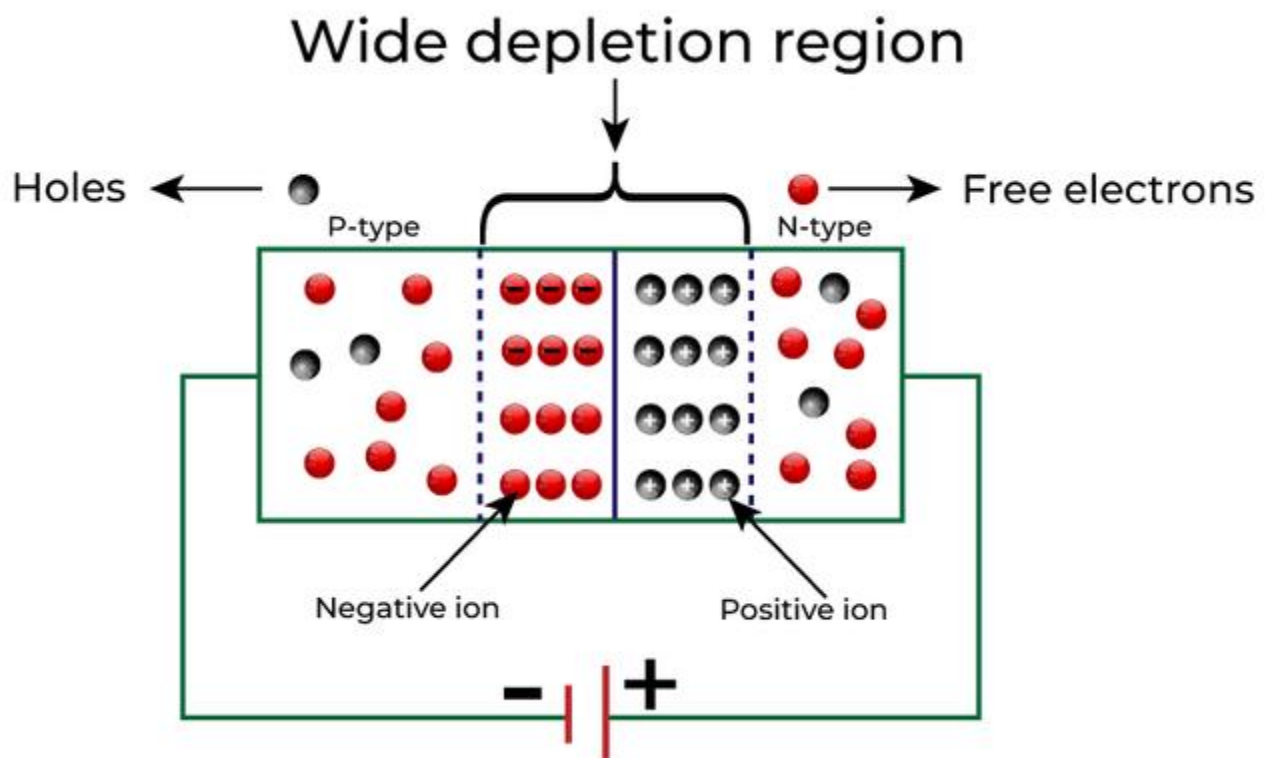
Working of Varactor diode

An important point to note is that varactor diode works in Reverse biased configuration.

Why does varactor diode work in reverse bias configuration ?

When a diode is subjected to forward bias, the majority carriers in both the p and n regions are pushed in the direction of the battery's applied voltage, leading to a reduction in the width of the depletion region. Eventually, the diode begins to conduct current more readily. However, a varactor diode is designed with a specific focus on its capacitance properties and its ability to store charges. This is why it is typically operated in reverse bias. When reverse biased, the varactor diode's capacitance varies with changes in the applied reverse voltage, making it useful in various applications like tuning circuits, frequency modulation, and voltage-controlled oscillators.

Let us understand how it works in reverse bias condition:-



reverse bias p-n junction diode

Initially, when there's no electric force applied, there's a thin empty region at the junction between two different materials in a diode. But when we apply a reverse electric force by connecting it to a battery the wrong way, something interesting happens.

The majority carriers in both the p-region (which has positive charges called "holes") and the n-region (which has negative charges called "electrons") start to move away from the junction. This means they get pushed apart because of the electric force. As a result, there are fewer of these majority carriers near the junction.

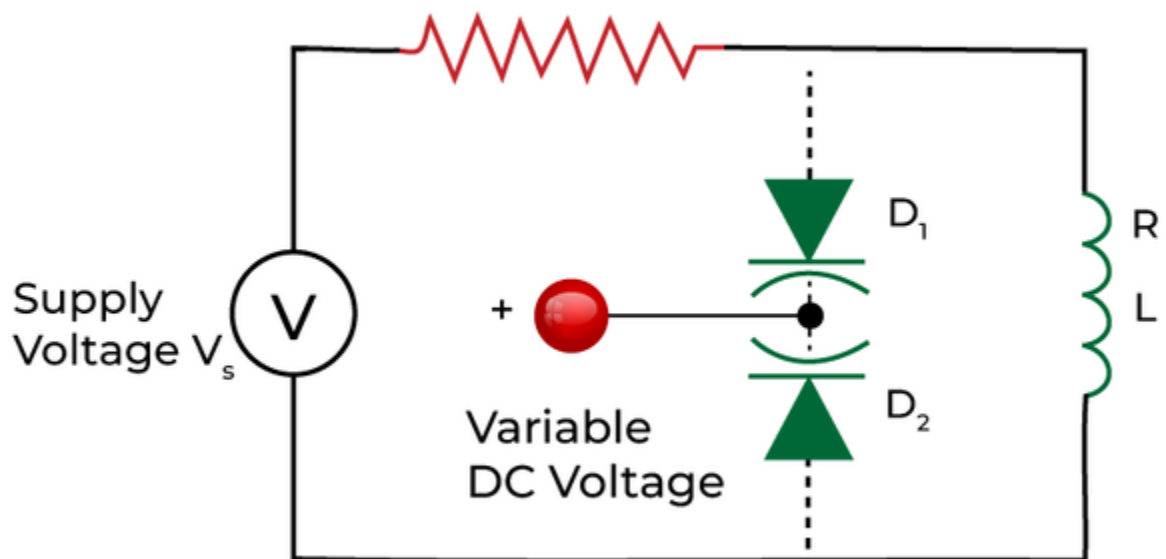
So, when we increase the reverse electric force by increasing the voltage from the battery, the empty region, which we call the "depletion region," gets bigger. It's like the gap between two plates of a capacitor. And the depletion region behaves like an insulating material in between those capacitor plates.

The capacitance at the junction is termed as Transition Capacitance.

Now, here's the important part: as we increase the reverse voltage (the electric force), the depletion region gets wider. And because of the formula for capacitance, when the width (W) of the depletion region increases, the capacitance (C_T) decreases.

So, in simple terms, when we change the voltage applied to the diode, the capacitance also changes. It gets smaller when we increase the voltage because the gap between the plates (the depletion region) gets bigger.

Varactor Diode in Tuning Circuit



varactor diode in tuning circuit

In this circuit

where,

L =Inductance in the Circuit(Henry)

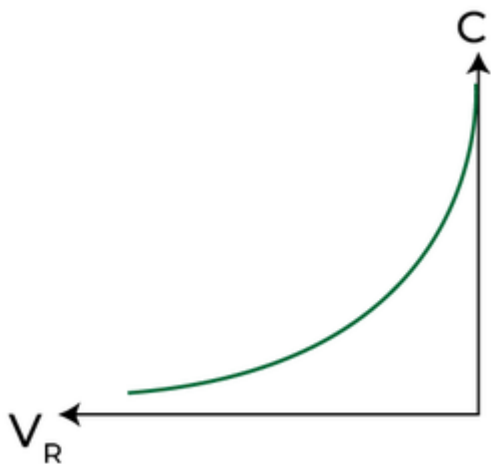
, C_{max} =Maximum Capacitance in diode

Varactor diodes are commonly used in tuning circuits, especially in radio frequency (RF) and microwave applications, to achieve variable capacitance and thus frequency tuning. Here's how varactor diodes are typically utilized in tuning circuits:

- A varactor diode is constructed similarly to a standard p-n junction diode, consisting of a p-type and an n-type semiconductor region separated by a depletion region. However, it's operated in reverse bias mode.
- When a reverse bias voltage is applied to the varactor diode (the p-side is connected to the negative terminal, and the n-side to the positive terminal of a power supply), it creates a depletion region around the p-n junction. This region is essentially a non-conductive area.
- The width of the depletion region in the varactor diode varies with the magnitude of the applied reverse voltage. As the reverse voltage increases, the depletion region widens, reducing the effective capacitance of the diode. Conversely, as the voltage decreases, the depletion region narrows, increasing the capacitance.
- In a tuning circuit, the varactor diode is connected in parallel with a fixed capacitor, often in an LC (inductor-capacitor) resonant circuit. By changing the reverse bias voltage across the varactor diode, its capacitance changes, which, in turn, alters the resonant frequency of the circuit.
- The ability to vary capacitance with voltage allows for precise and rapid frequency tuning. This makes varactor diodes valuable components in applications like voltage-controlled oscillators (VCOs), frequency synthesizers, and agile radio systems.

In summary, a varactor diode's ability to change its capacitance with applied voltage makes it a key component in tuning circuits.

Characteristic Curve of Varactor Diode



Characteristics curve of varactor diode

The graph shows the non-linear relationship between capacitance and voltage applied to diode. It is known that capacitance and width have an inverse relationship which means that as the width of the depletion region increases with the reverse voltage, the capacitance decreases (i.e. varies inversely as shown in the graph).

In short, as the reverse voltage increases, the transition capacitance falls rapidly. This behaviour can be describes as an exponential pattern.

Application of Varactor Diode

- **Voltage-Controlled Oscillators(VCOs):** Varactor Diodes are used in VCOs. VCOs are used in Phase-locked Loops and Communication Systems. By Varying the Voltage Across the Varactor Diode, the Capacitance Changes which leads to change of the Frequency in the Oscillator.
- **Frequency Modulation Tunning in Radios:** The Varactor Diodes are also used in the RF Circuits. By Varying the Voltage Across the Varactor Diode we can adjust the resonance Frequency of the tuned Circuit which we allow the tuning in FM radios.
- **Phase Shifters:** Varactor Diodes are also Implemented in Phase array Antennas. By Varying the Capacitance in the Certain part of the Antenna Array can lead to Controlling of the phase of the Emitted Signal Which can be used for beam Steering and Shaping.
- **Voltage-Controlled Filters:** Varactor Diodes are used in Voltage Controlled Bandpass or lowpass Filters. The Cutoff Frequency of the Filter can be Adjusted by Varying the Voltage across the varactor diode.

Advantages

Some common applications of varactor diode include:-

- **Variable Capacitance:** The main advantage of varactor diode is that we can use it as a variable capacitor. Due to this property it can be used as a part of frequency tuning circuits.
- **Frequency Synthesizer:** Due to the small size of varactor diode and variable capacitance, it can be used in high frequency elements of electronic devices to generate precise frequencies.
- **Phase Shifters:** In PLL(phase-locked loop) circuits, varactor diodes can be used to create voltage-controlled phase shifters, allowing precise phase control.
- **Frequency Multipliers and Dividers:** Varactor diodes are used in frequency multipliers and dividers to generate or divide frequencies as needed.
- **Economically Affordable:** Varactor diodes can be used in tuning circuits at many levels since they are economical and affordable. They also generate less noise as compared to other diodes.

Disadvantages

- **Mode-specific:** The designing of these diodes is done to operate them in reverse-bias hence they are not useful in forward bias.
- **Non-linear behaviour:** The major drawback of this diode is its non-linear capacitance-voltage relation which can result in distortion un many appliances.
- **Sensitive:** The capacitance of this diode is largely affected by temperature variations thereby making it temperature-sensitive.

Conclusion

In Conclusion, the varactor diode is PN Junction Diode which is Primarily Used for Reverse Bias for Variable Capacitance properties. The Characteristics Curve Shows a Non-Linear Relationship Between Capacitance and Voltage. Despite being Economically Affordable, Varactor Diodes have Limitations, Such as Mode-Specific Operations and Sensitive to temperature Variation. So In this article we have Gone Through Varactor Diode, we have seen all its working with Applications, Advantages and FAQs.